

Machine Learning Based Energy Demand Prediction

A Reproduction Study of Interpretable Demand-Response Baseline Models

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OBJECTIVE AND METHOD

Demand response payments are settled against a baseline — an estimate of what a site would have consumed had it not curtailed. Most markets compute this as a flat average of the same half-hour across recent eligible days, which ignores both the continuity of the load series and the weather. This project independently re-implements the framework of Kamoona et al. (2023), which argues that two textbook regression models beat that average while remaining simple, auditable and hard for participants to game — the properties a settlement baseline actually requires. Because the source data is proprietary, synthetic datasets were generated to the published structure: six C&I portfolios at 30-minute resolution over the availability windows of the original Table 1, each with one injected event day, and a single site of 48,288 observations from 01/04/2018 to 31/12/2020 with temperature, humidity, pressure, precipitation and solar radiation.

Predictors: the market average baseline; linear regression, $\hat{y} = \sum a_i x_i + b$; and second-order polynomial regression, $\hat{y} = \sum a_i x_i^2 + \sum b_i x_i + c$ (squared plus linear terms, no cross terms, so every coefficient remains readable). **Window geometries:** Model 1 uses the same clock position on the previous TW days (the baseline's own geometry); Model 2 uses the TW consecutive half-hours immediately preceding the target. **Metric:** RMSE, reported as a percentage of mean demand where portfolios of different scale are averaged. Implemented in Python (NumPy, pandas, scikit-learn, Matplotlib); seeded, deterministic, and runs end to end in under a minute.

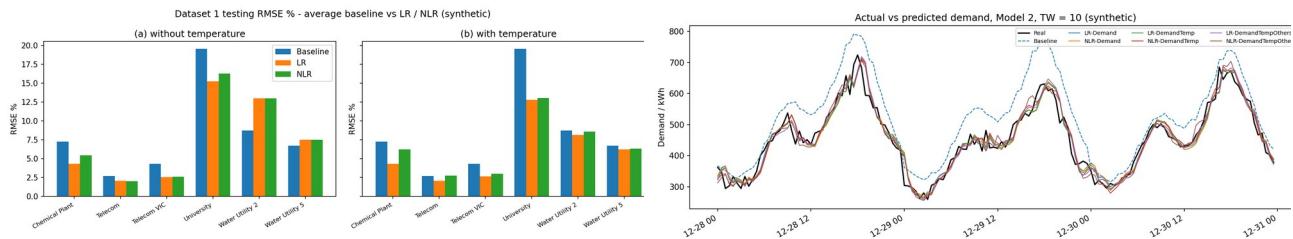
RESULTS

Case study 1 — six C&I portfolios (TW = 10)	Energy only	Energy + max temperature
Market average baseline	8.22 %	8.22 %
Linear regression (LR)	7.46 %	6.05 %
Polynomial regression (NLR)	7.82 %	6.66 %

Overall RMSE (%) averaged across the six portfolios. Both models beat the baseline; temperature improves both. Best result in bold.

Case study 2 — single site, test 28–30 Dec 2020 (RMSE, kWh)	TW = 4	TW = 7	TW = 10	TW = 14
Market average baseline	37.81	62.63	76.89	128.14
Model 1 (same-slot), best configuration	52.04	49.33	52.16	54.59
Model 2 (consecutive-lag), best configuration	30.72	28.51	28.11	27.78

Window geometry dominates: no Model 1 configuration falls below 49.3 kWh, while every Model 2 configuration lands between 27.8 and 31.4 kWh.



Left: per-portfolio RMSE (%) without (a) and with (b) temperature. Right: actual vs predicted demand over the test period, Model 2, TW = 10 — the dashed baseline overshoots each daily peak by 60–80 kWh while the regression traces are near-indistinguishable from the actual series.

FINDINGS

Reproduced. Both interpretable models beat the market baseline — overall on the six portfolios, and by a factor of more than four at TW = 10 on the single site. Lag-based windows beat same-slot windows universally, cutting error by roughly forty per cent regardless of model or feature set, confirming that respecting the continuity of the series is the single most valuable design choice available. Temperature improves every configuration tested; adding precipitation and solar radiation on top yields no reliable further gain, suggesting they are largely redundant with temperature at this resolution.

Diverged. On the six portfolios LR beats NLR (7.46 % vs 7.82 %), inverting the published ranking, and two portfolios are forecast worse by machine learning than by the flat average until temperature is added. Both are sample-size effects: half the portfolios supply under seven weeks of history, and NLR carries twice the coefficients for the same data. On the 48,288-observation single site the published ranking reappears, with NLR-DemandTemp best overall at 27.78 kWh. Absolute error levels are lower than published throughout because synthetic load is smoother than real metered consumption, which flatters the baseline specifically — making this a conservative test of the central claim rather than an optimistic one.

New to this reproduction. Window hygiene is not a detail. Because the polynomial model squares its inputs, an anomalous holiday or event day entering the averaging pool is amplified rather than diluted, and accuracy degrades sharply. Real settlement rules select the last TW *eligible* days for exactly this reason, and the implementation enforces the same filtering.

CONCLUSION

The accuracy available from replacing an averaging demand-response baseline is large and reachable with textbook methods. The two prerequisites are a lag-based input window and an aligned temperature feed; the choice between the linear and polynomial forms should be made on the length of available meter history rather than on general preference. The natural extension — and the one the source study nominates as its own future work — is to apply a tariff structure to the error reduction and express the improvement in currency rather than in kWh, which is the form in which the case would be put to a network operator or a customer.

Reference. Kamoona, A., Song, H., Keshavarzian, K., Levy, K., Jalili, M., Wilkinson, R., Yu, X., McGrath, B., & Meegahapola, L. (2023). Machine learning based energy demand prediction. Energy Reports, 9, 171–176. <https://doi.org/10.1016/j.egy.2023.09.151> — Independent reproduction study: the problem formulation, model definitions and experimental design are due to the authors above; the implementation, synthetic datasets, results, figures and analysis are the work of the present author.